Lesson plan · for the teacher

Logging data you can trust

IoT · Level 3 · Lesson 1

Time
40 min

Materials
A thermometer: a cheap dial or digital one is fine; A ruled log sheet and a pen by the door; Chalk or markers for sorting the readings; Optional: a phone with the free phyphox app, or a micro:bit (it has a temperature sensor)

Learning goals

- Plan a data log: what to measure, where, how often, with a time and unit on every reading.
- Check a sensor against a reference before trusting it.
- Clean glitches openly, and use the median when a few bad readings would drag the average.

Activity: A paper log on a rota

1. Warm-up (5 min): Write “Classroom: 85 °C at 7 am” on the board. Ask: could this be true, and what could cause it? Collect ideas, then explain the sensor that reads 85 °C as it powers up.
2. Plan (8 min): Agree one question with the class, such as “Is our room too hot after lunch?” Decide together what to measure, where to hang the thermometer (desk height, in the shade) and how often to read it. Rule a log sheet with columns for date, time, reading, unit and initials.
3. Check (7 min): Put two thermometers side by side for a few minutes. Pupils read both and write the readings in two columns. Is the difference steady? Explain an offset, and why we check before we trust.
4. Log (12 min): Set the rota so each pupil has an hour this week. Practise three readings now: eyes level with the scale, then the time, the number and the unit. Show how to write “missed” instead of leaving a blank.
5. Wrap-up (8 min): Give a sample afternoon: 31, 29, 85, 30, 28. Pupils sort it, find the median (30), work out the average with and without the 85, and write a one-line cleaning note.

Before class, hang the thermometer in the shade at desk height and check that pupils can read it. Use a dial or digital thermometer with young pupils; if a glass one breaks, an adult clears it up. If you use a sensor board, run it on batteries only.

Discussion

- Where else at school could a glitch mislead a decision?
- When should a surprising reading stay in the data?
- Who should be able to check how the data was cleaned?

Exit ticket

- Readings: 30, 28, 85, 31, 29 °C. What is the median, and what goes in the cleaning note?

Try it at school tomorrow

A paper log on a rota: Agree a question with your class, such as “Is our room too hot after lunch?” Hang a thermometer in the shade at desk height and set a rota: each hour, one pupil writes the time, the reading and the unit on a ruled sheet by the door. If you can, add a phone app such as phyphox, or a sensor board, beside it. At the end of the week, pupils sort the afternoon readings, find the median, mark any glitch and write a cleaning note.

Practice questions · for you

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1 **Zainab's pupils read the garden's soil moisture at noon each day: 62, 60, 0, 58, 0, 57 and 55 %. The two zeros came from a loose wire. Before any cleaning, which summary gives the fairest typical value?**

◯ The median ◯ The lowest reading ◯ The average

2 **A new sensor that always reads exactly 1 °C higher than a trusted thermometer is useless.**

◯ True ◯ False

3 **Put the steps of a trustworthy data log in order.**

☐ Log every reading with its time, unit and place

☐ Check the sensor beside a reference

☐ Copy the log, leave out glitches and note why

☐ Write down the question you want answered

☐ Summarise with a chart and the median

☐ Decide what to measure, where and how often

4 **In the Data logger lab, with "Needs water below" at 40 %, you leave out the four glitches (two readings of 0 and two of 100). Which numbers change a lot?**

☐ The highest reading ☐ The average ☐ The lowest reading ☐ The hours too dry

☐ The median

5 **Five afternoon readings came in: 31, 29, 85, 30 and 28 °C. Work out the summaries.**

Sorted, the middle reading is _____ °C, so that is the median. Leaving out the 85, the average of the other four is _____ °C.

Word bank: 40.6 · 85 · 30 · 29.5

✎ **Which question about your school could your pupils answer with a week of readings? What would they measure, where, and how often?**

Check your answers

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1 **The median**

Sorted, the readings are 0, 0, 55, 57, 58, 60, 62, so the median is 57 %, close to every real reading.

2 **False**

A steady difference is an offset: write it down and subtract 1 °C from every reading, and the sensor is as good as the reference.

3 **Top to bottom: 4, 3, 5, 1, 6, 2**

Each step rests on the one before: the question decides the measurement, a checked sensor makes the readings worth logging, and cleaning happens after logging, on a copy, so the raw data survives for anyone to check.

4 **The lowest reading; The hours too dry; The highest reading**

The glitches were exactly 0 and 100, so they set the lowest and highest values (about 55 and 85 without them) and faked two dry hours.

5 **Sorted, the middle reading is 30 °C, so that is the median. Leaving out the 85, the average of the other four is 29.5 °C.**

Sorted: 28, 29, 30, 31, 85, and the third of five is 30 °C.